

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (5 points) Find $\frac{dy}{dx}$ by implicit differentiation for the equation $\sin(x^3y^2) = x$.

Solution: $\frac{d}{dx} [\sin(x^3y^2)] = \frac{d}{dx} [x]$

$$\cos(x^3y^2) \frac{d}{dx} [x^3y^2] = 1$$

$$\cos(x^3y^2) [3x^2y^2 + 2x^3y \frac{dy}{dx}] = 1$$

$$2x^3y \frac{dy}{dx} = \frac{1}{\cos(x^3y^2)} - 3x^2y^2$$

$$\frac{dy}{dx} = \frac{1}{2x^3y \cos(x^3y^2)} - \frac{3y}{2x}$$

Problem 2. (5 points) Compute

$$\frac{d}{dx} \left[\ln \left(\frac{x^3 \cos x}{\sqrt{4-3x^2}} \right) \right]$$

Solution: $\ln \left(\frac{x^3 \cos x}{\sqrt{4-3x^2}} \right) = \ln x^3 + \ln \cos x - \ln \sqrt{4-3x^2}$

$$= 3 \ln x + \ln \cos x - \frac{1}{2} \ln (4-3x^2)$$

$$\Rightarrow \frac{d}{dx} \left[\ln \left(\frac{x^3 \cos x}{\sqrt{4-3x^2}} \right) \right] = \frac{d}{dx} \left[3 \ln x + \ln \cos x - \frac{1}{2} \ln (4-3x^2) \right]$$

$$= \frac{3}{x} + \frac{\sin x}{\cos x} - \frac{(-6x)}{2(4-3x^2)}$$

$$= \frac{3}{x} + \tan x + \frac{3x}{4-3x^2}$$